200G PAM4 DCI Architecture

A Deep Dive into 3nm CMOS SerDes Design, High-Speed Clocking, and Concatenated FEC Simulation

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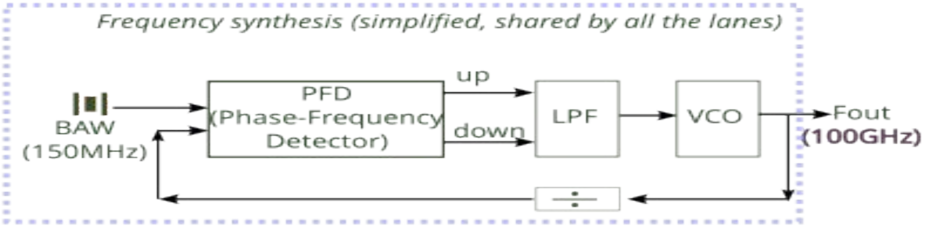
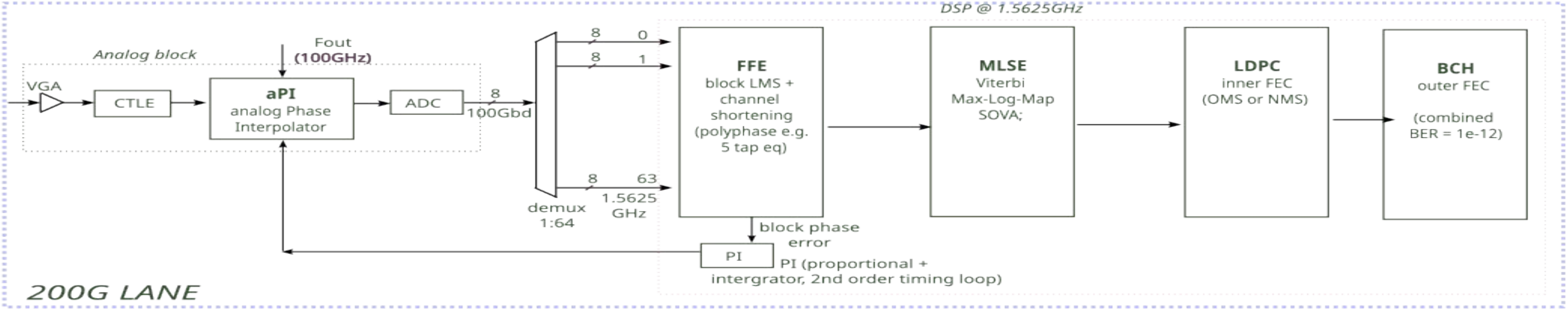
01. Lane Architecture


Frequency synthesis (simplified, shared by all the lanes)
BAW
(150MHz)
PFD
(Phase-Frequency
Detector)
up
down
LPF
VCO
Fout
(100GHz)


DSP @ 1.5625GHz
Analog block
Fout
(100GHz)
VGA
CTLE
aPI
analog Phase
Interpolator
ADC
8
100Gbd
demux
1:64
8
0
8
1
8
63
1.5625
GHz
FFE
block LMS +
channel
shortening
(polyphase e.g.
5 tap eq)
MLSE
Viterbi
Max-Log-Map
SOVA;
LDPC
inner FEC
(OMS or NMS)
BCH
outer FEC
(combined
BER = 1e-12)
block phase
error
PI
PI (proportional +
intergrator, 2nd order timing loop)
200G LANE

200G LANE

02. FEC Strategy

INNER CODE: LAYERED LDPC

40+

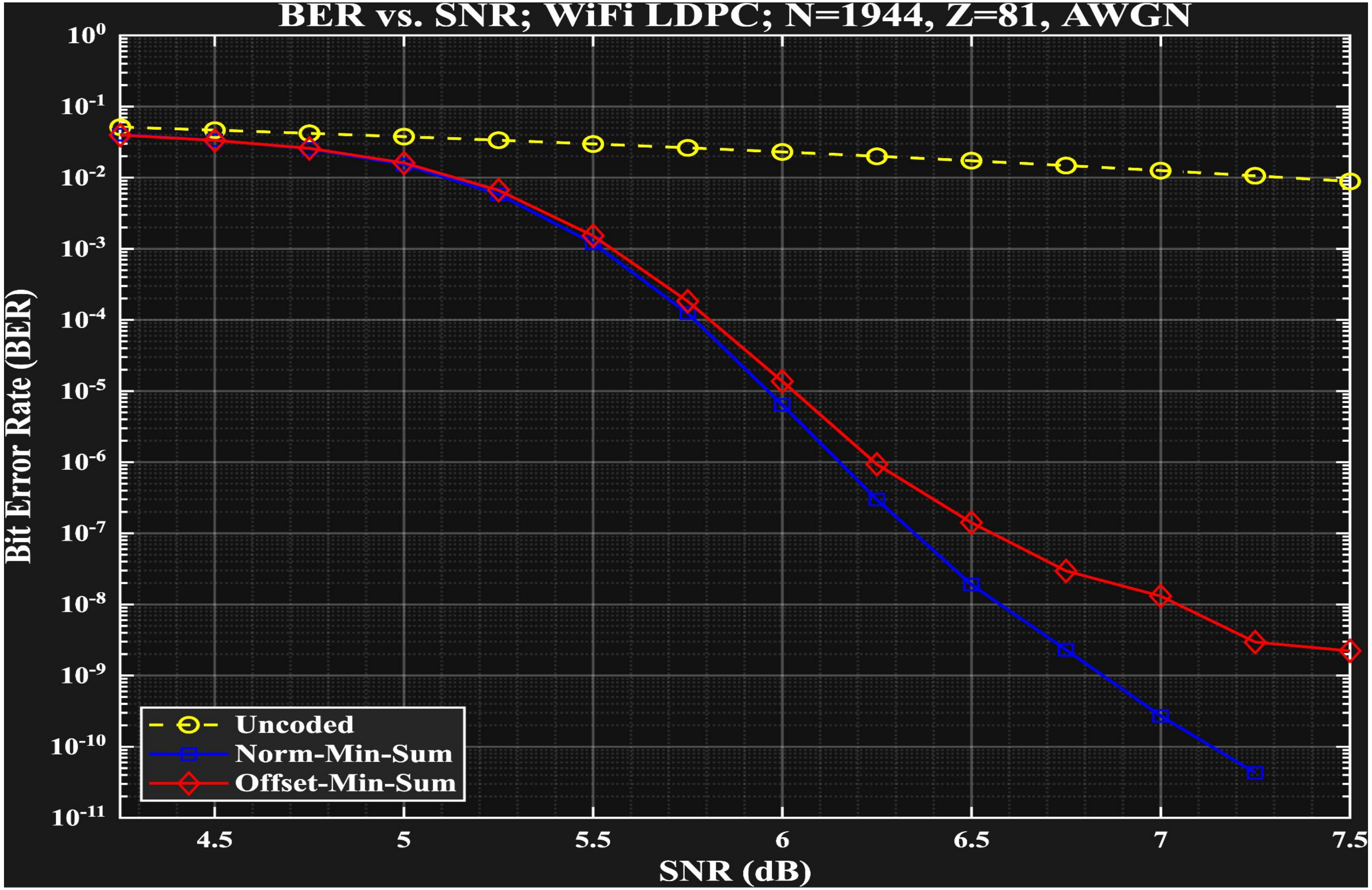
MBPS JAX THROUGHPUT

Soft-Decision Decoding

Inner Layered Belief Propagation (BP) decoder implemented in JAX/XLA. The model leverages **float16** state storage and **float32** compute to optimize RTX 3090 memory bandwidth.

$$Q_{v \to c} = L_v - R_{old, c \to v}$$

LDPC Reference BER vs SNR



OUTER CODE: BCH CLEANUP

- **Error Floor Mitigation:** Algebraic BCH decoding is used to clean up residual bit errors from LDPC "trapping sets" and "oscillations."
- **Low Latency:** BCH decoders provide deterministic latency, critical for DCI links that cannot afford iterative decoding variance.
- **Trapping Set Escape:** Quantization in JAX (float16) introduces numerical "regularization" that helps the decoder escape sub-optimal local minima.

CONCATENATED PERFORMANCE

Net Coding Gain

The combination of LDPC(1944, 5/6) and BCH(2048, 1944) delivers >11 dB NCG at 1e-12 BER floors.

JAX Scale-Up

Transitioning from 3090 single-GPU to A100 multi-GPU via jax.pmap allows simulation of 1e-15 BER targets.

MATLAB Verification

Perfect alignment with MATLAB 'approxllr' math verifies the PRNG and AWGN channel noise floor.

03. Timing Recovery

CONCLUSION & Q&A

3djax Project: GPU-Accelerated FEC Simulations.

Architecture: 3nm Monolithic 200G SerDes.

Innovation: JAX-based static graph optimization for high-speed hardware modeling.